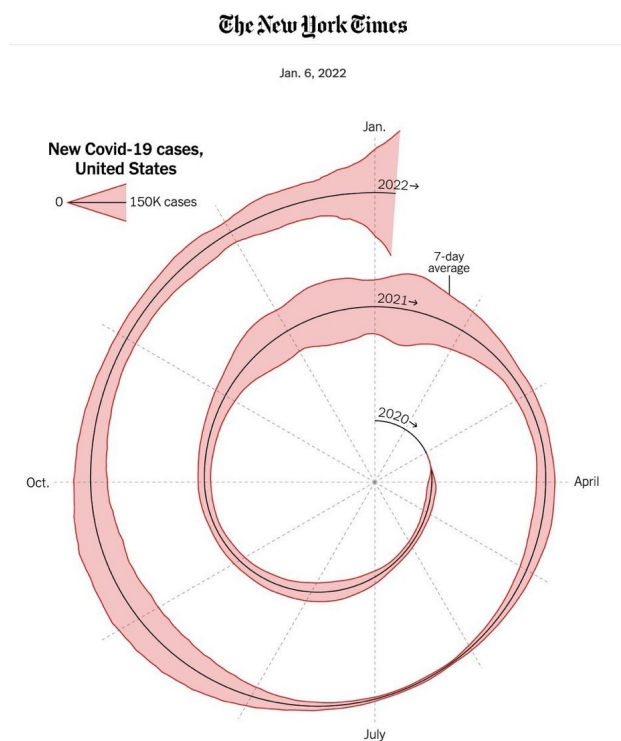


Data Visualization Assignment: Bad Visualization Analysis

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#1. The Bad Visualization Example I have selected a data visualization from The New York Times that illustrates Covid-19 cases in the United States using a spiral format.



Source: The New York Times, "Here's When We Expect Omicron to Peak", Jan 6, 2022.

Authors: Gus Wezerek & Sara Chodosh.

URL: <https://www.nytimes.com/2022/01/06/opinion/omicron-covid-us.html>

#2. The Problems Despite being visually striking, this spiral chart fails in several key data visualization principles:

Distorted Perception (Area Bias): The spiral layout distorts the visual weight of the data. As the spiral moves outward to later years (2022), the circumference increases. This causes the data points in the outer rings to occupy more physical space than those in the inner rings (2020), even if the numerical values are similar. This exaggerates the visual impact of recent data.

Difficulty in Comparison: It is cognitively difficult to compare the magnitude of peaks (wave heights) across different years because the baseline is curved and rotating. The viewer cannot easily draw a straight line to compare the Winter 2020 peak with the Winter 2021 peak.

Non-Intuitive Axis: Humans naturally perceive time linearly (from left to right). Wrapping the timeline into a spiral increases cognitive load, forcing the reader to mentally “unwind” the plot to understand the chronological progression.

#3. The Solution To solve these problems, the data should be plotted on a standard Cartesian coordinate system using a Linear Line Chart or an Area Chart.

X-Axis: Should represent Time linearly.

Y-Axis: Should represent the Number of Cases.

This approach removes the radial distortion and allows for an accurate, direct comparison of the wave magnitudes.